Lab 7 — Fuzz Testing with Hypothesis

Blood Donation Screening | Student: Tejas Shetty | PES University

Objective

This report documents the comparison between the original input-processing implementation and the corrected implementation using the Lab 7 fuzz/property-based test suite.

Functions tested

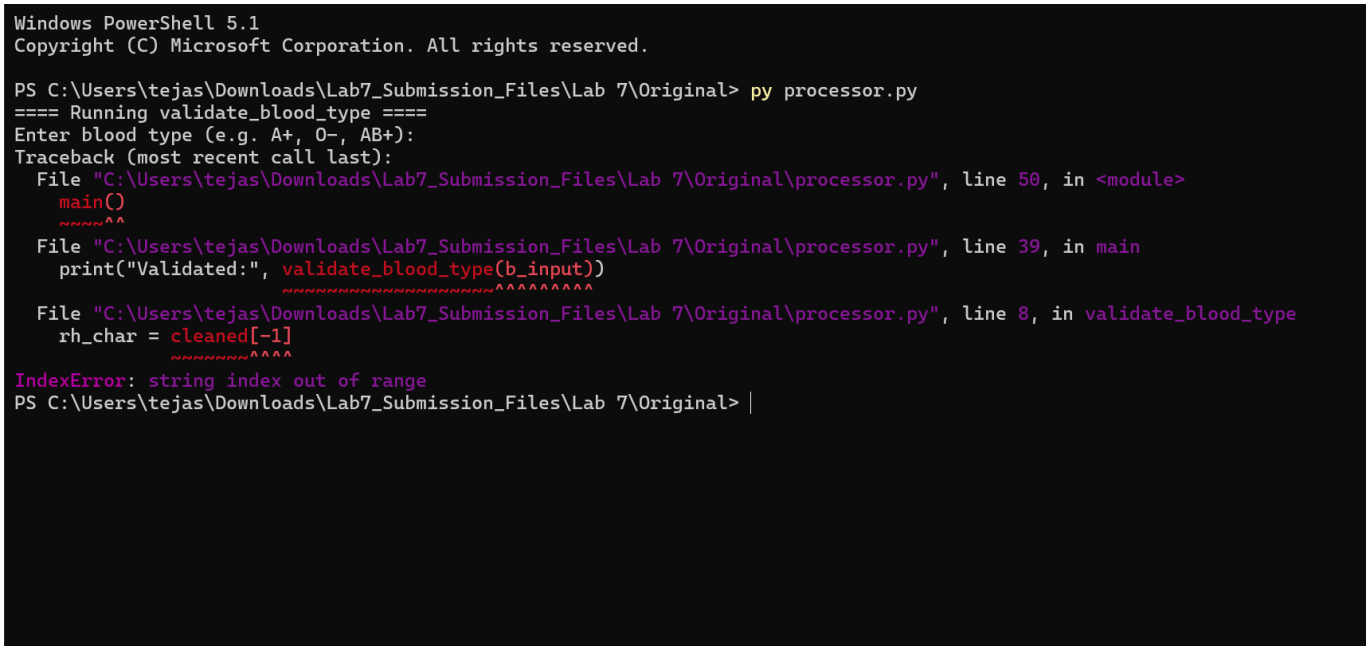
• validate_blood_type() — ABO/Rh blood-type validation
• calculate_pulse_pressure() — systolic/diastolic parsing and metric calculation
• parse_donor_registry() — donor-record parsing and eligibility decision

Execution summary

The original program was observed to fail on empty blood-type input. The fuzz tests then exposed additional invalid-input handling defects in the original implementation. After applying input validation and error handling, the application produced the expected sample outputs and the test run completed with 33 passing tests.

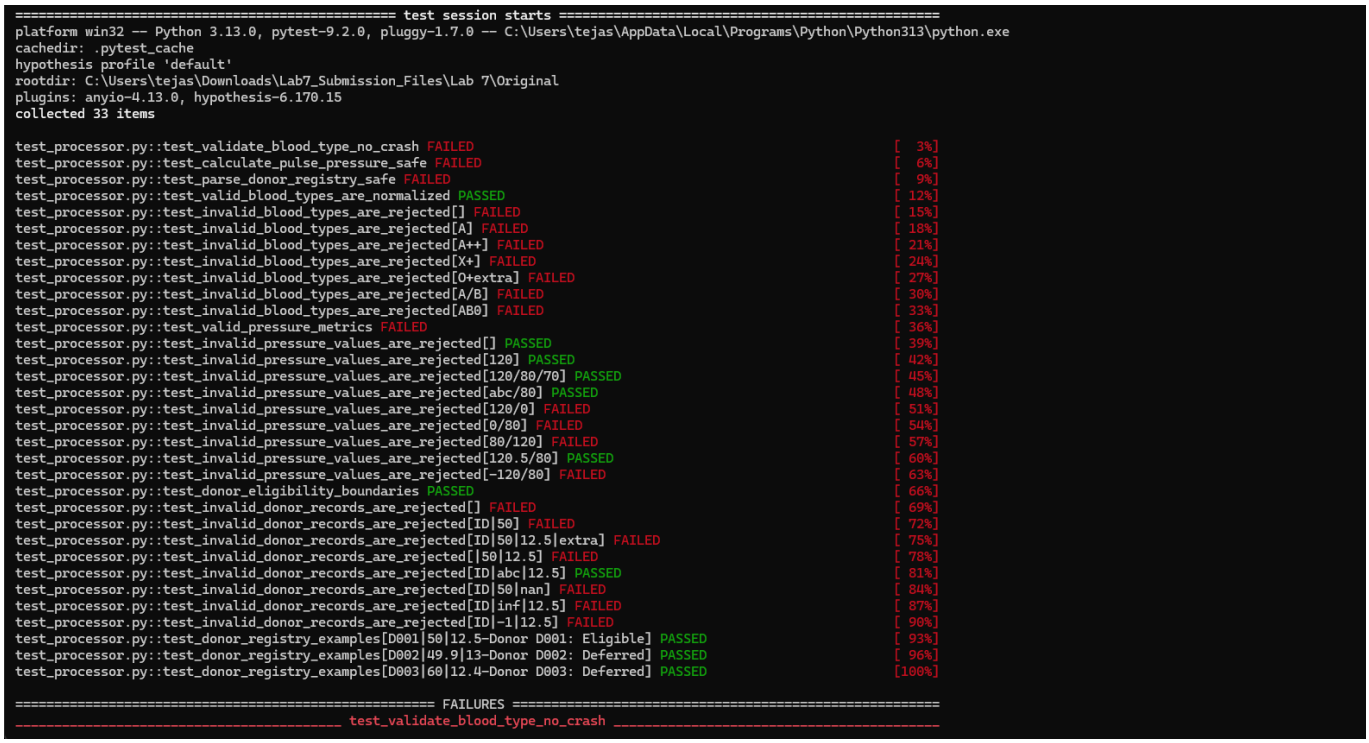
Evidence 1 — Original buggy application

The original processor crashes on an empty blood-type input with IndexError: string index out of range.



Evidence 2 — Fuzz tests against the original implementation

The completed test suite reports failures across blood-type validation, blood-pressure parsing, and donor-registry parsing.



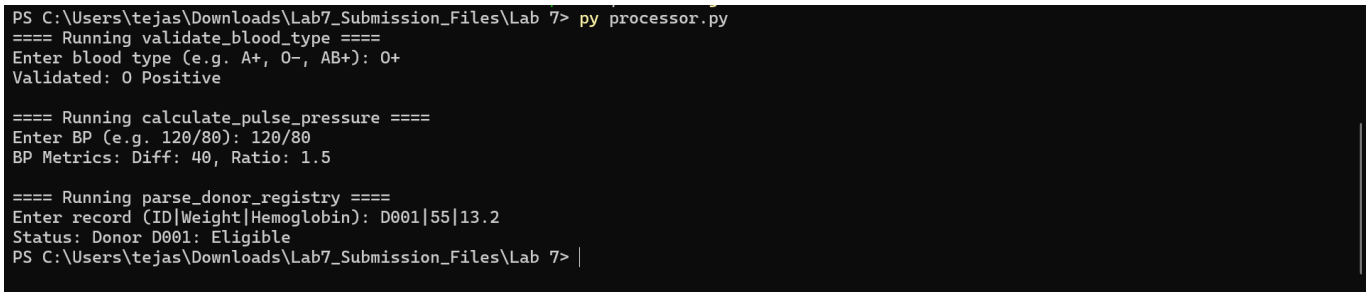
```
============================================= test session starts =============================================
platform win32 -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\tejas\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\tejas\Downloads\Lab7_Submission_Files\Lab 7\Original
plugins: anyio-4.13.0, hypothesis-6.170.15
collected 33 items

test_processor.py::test_validate_blood_type_no_crash FAILED                                          [  3%]
test_processor.py::test_calculate_pulse_pressure_safe FAILED                                         [  6%]
test_processor.py::test_parse_donor_registry_safe FAILED                                             [  9%]
test_processor.py::test_valid_blood_types_are_normalized PASSED                                      [ 12%]
test_processor.py::test_invalid_blood_types_are_rejected[] FAILED                                    [ 15%]
test_processor.py::test_invalid_blood_types_are_rejected[A] FAILED                                   [ 18%]
test_processor.py::test_invalid_blood_types_are_rejected[A++] FAILED                                 [ 21%]
test_processor.py::test_invalid_blood_types_are_rejected[X+] FAILED                                  [ 24%]
test_processor.py::test_invalid_blood_types_are_rejected[O+extra] FAILED                             [ 27%]
test_processor.py::test_invalid_blood_types_are_rejected[A/B] FAILED                                 [ 30%]
test_processor.py::test_invalid_blood_types_are_rejected[AB0] FAILED                                 [ 33%]
test_processor.py::test_valid_pressure_metrics FAILED                                                [ 36%]
test_processor.py::test_invalid_pressure_values_are_rejected[] PASSED                                [ 39%]
test_processor.py::test_invalid_pressure_values_are_rejected[120] PASSED                             [ 42%]
test_processor.py::test_invalid_pressure_values_are_rejected[120/80/70] PASSED                       [ 45%]
test_processor.py::test_invalid_pressure_values_are_rejected[abc/80] PASSED                          [ 48%]
test_processor.py::test_invalid_pressure_values_are_rejected[120/0] FAILED                           [ 51%]
test_processor.py::test_invalid_pressure_values_are_rejected[0/80] FAILED                            [ 54%]
test_processor.py::test_invalid_pressure_values_are_rejected[80/120] FAILED                          [ 57%]
test_processor.py::test_invalid_pressure_values_are_rejected[120.5/80] PASSED                        [ 60%]
test_processor.py::test_invalid_pressure_values_are_rejected[-120/80] FAILED                         [ 63%]
test_processor.py::test_donor_eligibility_boundaries PASSED                                          [ 66%]
test_processor.py::test_invalid_donor_records_are_rejected[] FAILED                                  [ 69%]
test_processor.py::test_invalid_donor_records_are_rejected[ID|50] FAILED                             [ 72%]
test_processor.py::test_invalid_donor_records_are_rejected[ID|50|12.5|extra] FAILED                  [ 75%]
test_processor.py::test_invalid_donor_records_are_rejected[|50|12.5] FAILED                          [ 78%]
test_processor.py::test_invalid_donor_records_are_rejected[ID|abc|12.5] PASSED                       [ 81%]
test_processor.py::test_invalid_donor_records_are_rejected[ID|50|nan] FAILED                         [ 84%]
test_processor.py::test_invalid_donor_records_are_rejected[ID|inf|12.5] FAILED                       [ 87%]
test_processor.py::test_invalid_donor_records_are_rejected[ID|-1|12.5] FAILED                        [ 90%]
test_processor.py::test_donor_registry_examples[D001|50|12.5-Donor D001: Eligible] PASSED            [ 93%]
test_processor.py::test_donor_registry_examples[D002|49.9|13-Donor D002: Deferred] PASSED            [ 96%]
test_processor.py::test_donor_registry_examples[D003|60|12.4-Donor D003: Deferred] PASSED            [100%]

================================================== FAILURES ==================================================
_____________________________________ test_validate_blood_type_no_crash _____________________________________
```

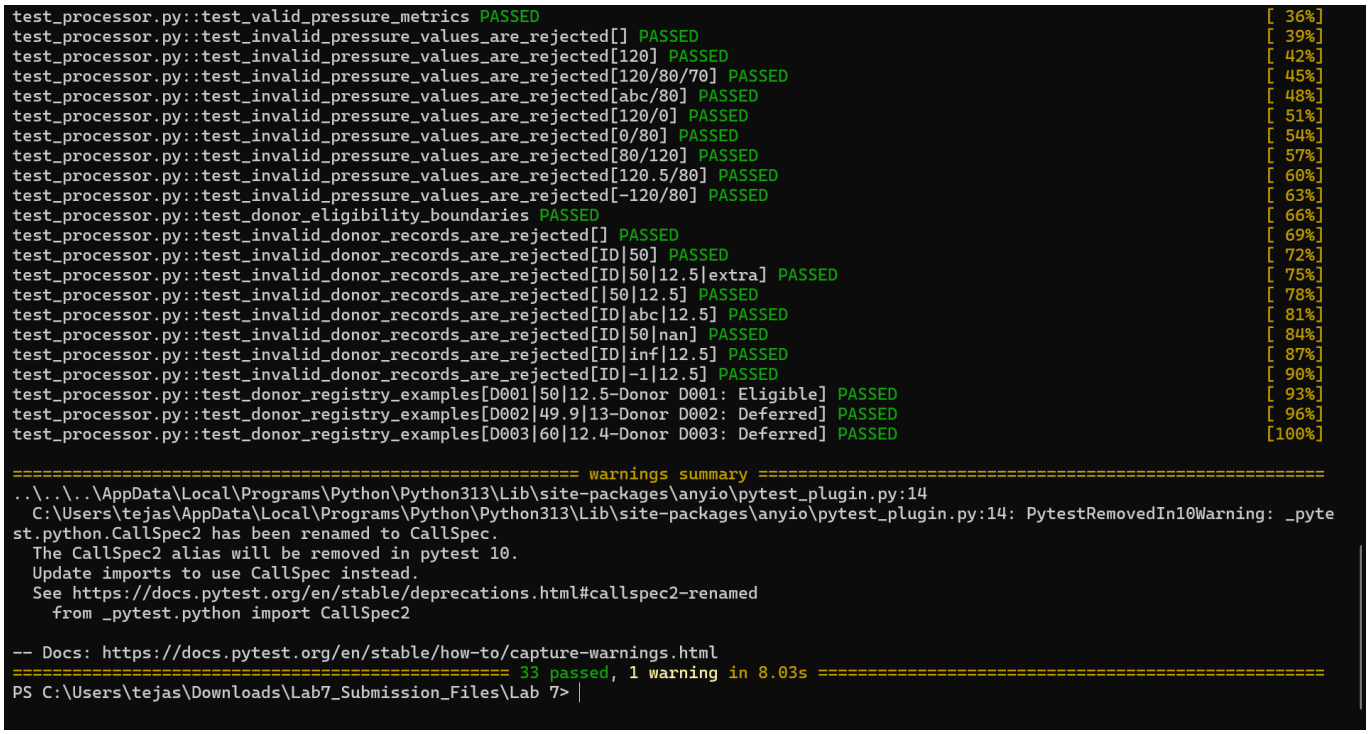
Evidence 3 — Fixed application execution

The corrected application processes O+, 120/80, and D001|55|13.2 and returns the expected outputs.



Evidence 4 — Fuzz tests against the fixed implementation

The completed test suite finishes with 33 passed and one non-blocking deprecation warning.



Conclusion

The captured runs demonstrate the intended fuzz-testing workflow: observe defects in the original implementation, use generated and targeted inputs to expose failures, correct the input handling, and rerun the tests to verify the fixes. The final local test run shown in the evidence reports 33 passed. One pytest/AnyIO deprecation warning was also reported; it did not cause test failure.

Submission files: processor.py, test_processor.py, and this PDF report. Include requirements.txt if your instructor expects dependency instructions.